

Subject Description Form

Subject Code	AMA541
Subject Title	Simulation and Risk Analysis
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Enable students to appreciate the principles and methods of system simulation. Emphasis is placed on the process of translating real-world financial and risk problems into simulation models, and the model building techniques involved.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Explain the basic concepts of simulation and its utility in solving real-world financial and risk problems. (b) Apply mathematical and statistical knowledge and modelling techniques required to construct simulation models for real-world systems. (c) Apply statistical knowledge and techniques to verify and validate simulation models. (d) Analyze and interpret simulation outputs. (e) Present results of simulation analysis.
Subject Synopsis/ Indicative Syllabus	Fundamental of Simulation Models: Principles of mathematical simulation, advantages and disadvantages of simulation, types of simulation models, steps in a simulation study. Discrete-Event Simulation: General principles, components and organization of a discrete-event simulation model, simulation examples (e.g. queueing and inventory systems, financial and risk models), event scheduling, gathering summary statistics. Random Numbers and Random Variates: Generation of pseudo-random numbers, Statistical tests of randomness, generation of random variates, and applications in simple simulation experiments. Monte Carlo Simulation and Stochastic Models: Concepts and methodology of Monte Carlo simulation, with applications in financial modeling and risk analysis. Introduction to stochastic processes in finance, including models for asset price dynamics and related simulation approaches. Financial Risk Applications: Simulation methods applied to financial decision making and risk management. Topics cover portfolio modeling, risk measurement techniques, and credit risk analysis, with an emphasis on quantitative approaches and data-driven applications.

	Verification and Validation: Verification and validation of simulation models, comparisons, appropriate statistical tests, sensitivity analysis, simulation run statistics, replication of runs, elimination of initial bias, batch means, and regenerative techniques. Computer Language for Simulation: Python																																													
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The teaching and learning approach is mainly problem-solving oriented. The approach aims at the development of mathematical and statistical techniques and how the techniques can be applied to solving problems. Students are encouraged to adopt a deep study approach by employing high level cognitive strategies, such as critical and evaluative thinking, relating, integrating and applying theories to practice.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignments/Projects</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Tests</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Examination</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> (a), (b), (c) and (d) will be assessed by assignment/test, case study/project and final examination. (e) will be assessed by case study/project. Typical case study/project includes inventory management, queueing system, financial model, and/or risk management models.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignments/Projects	20%	✓	✓	✓	✓	✓	2. Tests	40%	✓	✓	✓	✓	✓	3. Examination	40%	✓	✓	✓	✓	✓	Total	100 %					
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Student Study Effort Required	<table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Lecture</td><td>26 Hrs.</td></tr><tr><td>▪ Tutorial</td><td>13 Hrs.</td></tr><tr><td>Other student study effort:</td><td></td></tr><tr><td>▪ Assignment/Project</td><td>58 Hrs.</td></tr><tr><td>▪ Self-study</td><td>40 Hrs.</td></tr><tr><td>Total student study effort</td><td>137 Hrs.</td></tr></table>						Class contact:		▪ Lecture	26 Hrs.	▪ Tutorial	13 Hrs.	Other student study effort:		▪ Assignment/Project	58 Hrs.	▪ Self-study	40 Hrs.	Total student study effort	137 Hrs.																										
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		3 rd Edition	
	Ross, S.M.	Simulation, 5 th Edition	Academic Press, 2013
	Ross, S.M.	A Course in Simulation	MacMillan, 1991
	Selia, A.F., Ceric, V. and Tadikamalla, P.	Applied Simulation Modeling	Thomson, Brooks/Cole, 2003